

HANDIKO GESANG ANUGRAH SEJATI

Jakarta | 085741249084 | handikogesang@gmail.com | Portfolio: handiko.github.io/algo

PROFESSIONAL SUMMARY

Quantitative-focused Algorithmic Trader and Developer with a deep understanding of market microstructure across Crypto, Forex, and Equities. Specialized in developing automated execution systems in MQL5, with a strong emphasis on liquidity analysis, risk mitigation, and order book dynamics. Seeking to transition technical expertise in market mechanics into the Crypto Business Operations to optimize exchange liquidity and execution quality.

TECHNICAL SKILLS

- **Market Operation:** Order Book Management, Slippage Analysis, Market Maker Performance Tracking, Liquidity Provision.
 - **Market Analysis:** Order Book Dynamics, Liquidity Mechanics, Bid-Ask Spread Optimization, Hedging.
 - **Risk Management:** Inventory Risk Modelling (Avellaneda-Stoikov model), Anomaly Detection.
 - **Trading Systems:** MQL5 (Expert Advisor), MetaTrader 5, Algorithmic Strategy Design.
 - **Data & Tools:** MQL5 Backtesting & Optimization, Risk Management Modules (Python), Financial data analysis (Python).
 - **Core Competencies:** P&L Tracking, Market Anomaly Detection, Quantitative Research.
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RELEVANT EXPERIENCE

Independent Algorithmic Trader & Developer | 2023 – Present

- Developed and deployed market maker monitoring and simulation dashboard using Avellaneda-Stoikov model (<https://handiko.github.io/market-maker-sim/>).
- Developed and deployed custom automated trading systems in MQL5 for Forex and Crypto markets (Portfolio: handiko.github.io/algo).

- Analyzed market microstructure to optimize entry/exit logic, focusing on minimizing slippage and maximizing execution quality in low-liquidity environments.
- Managed a diversified portfolio across Crypto, Forex, and Stocks, maintaining consistent P&L tracking and performance attribution.
- Conducted extensive backtesting and forward-testing to identify pricing inefficiencies and market anomalies.

SELECTED PROJECTS

Avellaneda-Stoikov Market Making Simulation

- The simulator models a market maker (MM) who provides liquidity by simultaneously posting bid and ask quotes. The MM's goal is to capture the spread while managing **Inventory Risk** (the risk of price movement against a held position) and **Adverse Selection** (the risk of trading against informed participants).
- Live page: <https://handiko.github.io/market-maker-sim/>

USDT-IDR Market Monitor

- A high-fidelity monitoring terminal for the USDT/IDR pair, built for liquidity analysis and execution simulation across Indonesian exchanges. This project focuses on minimizing latency and maximizing the visibility of order book depth.
- Page: <https://handiko.github.io/USDT-IDR-Monitoring-Dashboard/>

Treasury Monitoring Terminal: Smart Rebalancing Manual

- Treasury monitoring terminal simulation app to demonstrate the treasury rebalancing process in a crypto exchange. A simplified example.
- In the ecosystem of a crypto exchange, “Inventory Management” is a critical pillar of Business Operations. Capital is generally split between two venues: Local Hot Wallets and Global Liquidity Pools.
- If a major market event triggers a “Bank Run” (mass withdrawals), the Local Wallet can be depleted in minutes. Conversely, keeping too much capital locally increases security risks and incurs opportunity costs.
- This terminal acts as a decision-support system that monitors these levels in real-time. It uses a **Drift-Band Strategy** to ensure the exchange stays solvent while minimizing transaction friction and network fees.

- Page: <https://handiko.github.io/treasury-terminal/>

MOST RECENT WORK

PT. Imani Prima | *Firmware Engineer* | **2020 – 2025**

- Managed R&D for AIS (Automatic Identification System) transmitters and receivers used by national maritime and fisheries institutions projects, requiring high attention to detail and data-driven decision-making.
 - Collaborated with cross-functional teams to improve operational workflows.
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EDUCATION

Universitas Gadjah Mada (UGM) | *Bachelor of Engineering in Engineering Physics*

- **Specialization:** Instrumentation and Control.
 - **Final Project:** Multi-channel modulators for Software Defined Radio.
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